Professional Issues in IT

MANAGEMENT OF ACCOUNTING

Management Accounting

After reading this chapter, you should, in simple cases:

be able to calculate the cost of labour;

understand the concept of overheads and the different ways in which they may be distributed;

be able to determine how much it costs to produce a particular product or provide a specific service;

understand how to produce a budget and a cash flow forecast, and how to monitor them

COST OF LABOUR

- The cost of employing someone is more than just the cost of their salary.
- In most countries, employers are required to pay a tax for every employee. This tax usually goes by a name such as *employers' national insurance contribution* (in the UK) or *social security contribution; it is proportional to the* employee's salary. In some countries, this contribution may be as large as 60 per cent of salary, while in others it is very much smaller.
- The total cost of employing a person, that is, the salary plus employers' social security contributions plus any other costs associated directly with the employee, is sometimes known as the **employee's payroll cost or direct cost.**

Cost of labour

TABLE 7.1 *Calculation of the number of revenue-earning hours in a year*

Total number of weekdays (1)	260
Public holidays (2)	10
Annual leave (3)	20
Sick leave (4)	5
Unproductive time (5)	10
Total non-revenue-earning time (2) + (3) + (4) + (5) = (6)	45
Total number of revenue earning days (1) – (5) = (7)	215
Total number of hours available (7) × 7	1505

Pay roll

- The total cost of employing a person, that is, the salary plus employers' social security contributions plus any other costs associated directly with the employee, is sometimes known as the employee's *payroll cost or direct cost.*

Example :

- *Consider a technician who is employed* to assemble the computers and suppose that he is paid an annual salary of £20,000. His payroll costs are £22,000.
- The direct cost of the technician is £22,000 per year so that the direct cost of an hour of their time is £22000/1505 = £14.62.

Example

- Employees of Syniad Software plc work a five day week. They are entitled to 20 days holiday a year in addition to public holi days. On average, each employee loses 10 working days per year through sickness. Syniad aims to allow each employee 15 days per year for training. Experience shows that employees spend an aver age of five days a year unproductively, as a result of scheduling problems. In accordance with government regulations, employers must pay social security contributions equal to 6 per cent of salary. 83 Professional Issues in Information Technology Calculate the average direct cost of one day's work and hourly cost of labour from an employee earning £20,000 per year. State explicitly any assumptions you make.

Example

- The employer's contribution is equal to 3% of an employee's salary. The employer also has to pay what is called the Employer's National Insurance contribution. At the time of writing, this is 13.8% of salary above £9,564 per year.
- Calculate hourly direct cost.

OVERHEADS

- Costs that cannot be directly associated with a particular product are known as overheads.
- In a large business, there are many ways in which overheads can be spread over the different activities the business performs; there is no single ‘right’ way of doing this.
- If the components of a computer cost £200 and it takes 10 hours of a technician’s time to assemble the computer, install the software and configure it, then the cost of constructing the finished product is £200 + 10 × £14.62 = £346.20.

Overheads

TABLE 7.2 *Direct costs and expected sales for the different computer models*

Model	Cost of components (£)	Technician time (hours	Expected sales
Basic	£200	10 hours	200
Advanced	£300	12 hours	100
Professional	£400	15 hours	50

Allocation of Overheads

- The company has to recover the overhead costs of **£63,500.**
- The only way it can do this is collect it from the income it gets from sales. However, there are many different ways in which this can be done. We shall describe three of the most common ways of doing this.

Allocation of Overheads

TABLE 7.3 *Effects of different overhead calculations*

	Fixed overhead	Overhead proportional to labour content	Overhead proportional to total direct cost
Basic	528	487	495
Advanced	652	644	680
Professional	802	830	886

BUDGETING

- A budget is a financial plan showing the expected income and expenditure for an organization over a specific period, typically one year.
- In order to produce a budget for a company, we have to make some assumptions about what it will be doing.

Example

TABLE 7.4 *An example budget*

Overhead Expenditure	
Owner's payroll costs	42,000
Secretary's payroll costs (part time)	8,000
Costs of van (including depreciation)	3,500
Internet connection, telephone, postage, etc.	1,000
Advertising	2,000
Premises (heating and lighting, rent, rates, etc.)	4,500
Professional fees	1,000
Insurance	500
Total overheads	63,500
Operating costs	
Technicians' payroll costs	66,000
Bought-in components	90,000
Total manufacturing costs	156,000
Total costs	219,500
Sales income	
Basic model (200 @ £595)	119,000
Advanced model (100 @ £795)	79,500
Professional model (50 @ £895)	44,750
Total sales	243,250
Profit	24,750

BUDGETING

- The costs under the heading ‘overhead expenditure’ are ones that the company will have to pay regardless of how many computers it sells. For this reason, they are often known as fixed costs.
- The figures under the heading ‘operating costs’ depend on the number of computers the company sells. For this reason, they are known as variable costs. The cost of components is directly connected to the number of computers sold.

BUDGETING

- Once a budget has been agreed, it should be used to monitor the company's financial progress. The first step is to break it down to show monthly income and expenditure – a budget broken down in this way is sometimes called a **profiled budget.**

CASH FLOW FORECAST

- A company may be very profitable but unable to pay its bills. For that reason, it may be forced into receivership. This apparent paradox typically arises because bills have to be met, in particular staff have to be paid, before the income they generate is received.
- In order to avoid this difficulty businesses need to prepare cash flow forecasts, that is, estimates of the amount of money that will flow into and out of the company each month.

Month	Jan	Feb	March	April	May	June
Cash outflow						
Rent and property taxes	500			500		
Energy costs		400	400	300	200	200
Payroll costs	9,666	9,666	9,666	9,666	9,666	9,666
Communications		83	83	83	83	83
Insurance	500					
Components		4,000	7,000	10,000	10,000	10,000
Advertising		500		250		500
Road tax and insurance on van	700					
Professional fees			300			
Van operating costs	100	100	100	100	100	100
Monthly cash outflow	11,466	15,349	17,549	20,899	20,049	20,549
Cash inflow						
Income from retail sales	5,000	5,000	5,000	5,000	5,000	5,000
Income from trade sales		5,000	7,000	10,000	15,000	18,000
Monthly cash inflow	5,000	10,000	12,000	15,000	20,000	23,000
Net monthly cash flow	(6,466)	(5,349)	(5,549)	(5,899)	(49)	2,451
Cumulative cash flow	(6,466)	(11,815)	(17,364)	(23,263)	(23,312)	(22,861)

Budget and CASH FLOW FORECAST

- At first sight, cash flow forecasts and budgets seem very much the same thing. It is important to understand the difference. Cash flow forecasts deal with the flow of cash or its equivalents in and out of the company. Budgets deal with income and expenditure.
- A **cash flow forecast** is essential for managing the **timing** of cash inflows and outflows, ensuring the business stays liquid.
- A **budget** is a strategic tool used to plan revenues and expenses over a period, focusing on achieving **profitability goals**.